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INDUSTRIAL VISIT REPORT



"AIRAA Engineering & Kiruthika Industries"

For

STUDENTS INNOVATION CLUB

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INDUSTRIAL VISIT REPORT 1 -
AIRAA ENGINEERING, COIMBATORE

Host Company	AIRAA Engineering, Coimbatore, Tamil Nadu
Date of Visit	14 August 2026 (Friday – FN session)
Department	Mechanical Engineering
Focus Areas	Fiber laser cutting, MIG welding, sheet metal fabrication

1. Introduction

As part of our course work, our team went to AIRAA Engineering in Coimbatore for an industrial visit. It's a sheet-metal fabrication unit, and the idea was to see how the things we've been reading about in Manufacturing Processes, Materials, and CAD/CAM classes actually happen on a shop floor.

The main things we got to see were **fiber laser cutting** and **MIG welding**. We watched a plain sheet of metal get turned into a finished, cut-to-shape component right in front of us, and the staff there walked us through the materials, gases, and software that go into it.

2. About AIRAA Engineering

AIRAA Engineering is a fabrication and job-work unit based in Coimbatore. They mainly do CNC **fiber laser cutting** and **MIG welding** for other companies, basically, customers send drawings and they cut the components to size in mild steel, stainless steel, GI, and aluminum.

3. Objectives of the Visit

- Understand how **fiber laser cutting** actually works, not just in theory.
- See **MIG welding** being done and how it's used in fabrication work.
- Learn about the different sheet materials, MS, GI, SS, aluminum, and how they're processed.
- Find out why different gases (**oxygen** and **nitrogen**) are used for different metals.
- Understand the difference between hot-rolled and cold-rolled mild steel.
- Get a look at the CNC control panel and the cutting software.
- Understand why a stable power supply matters so much for a machine like this.
- See how a CAD drawing eventually becomes a cutting program on the machine.

4. Laser Cutting Process

This was the main thing we went to see. AIRAA Engineering runs a **3 kW fiber laser cutting** machine that cuts flat sheet metal into shapes based on customer drawings. LASER, as our guide reminded us, stands for **Light Amplification by Stimulated Emission of Radiation**, the machine focuses a very intense beam onto the sheet, which melts and burns through the metal along the programmed path, giving a clean, narrow cut.

The machine has a **CNC control unit** with a touchscreen, a keyboard, and a big red **emergency-stop** button (you can see it in Fig. 1 and Fig. 3). From there the operator loads the cutting file, watches the cut happen live, and can tweak the speed or power if needed.

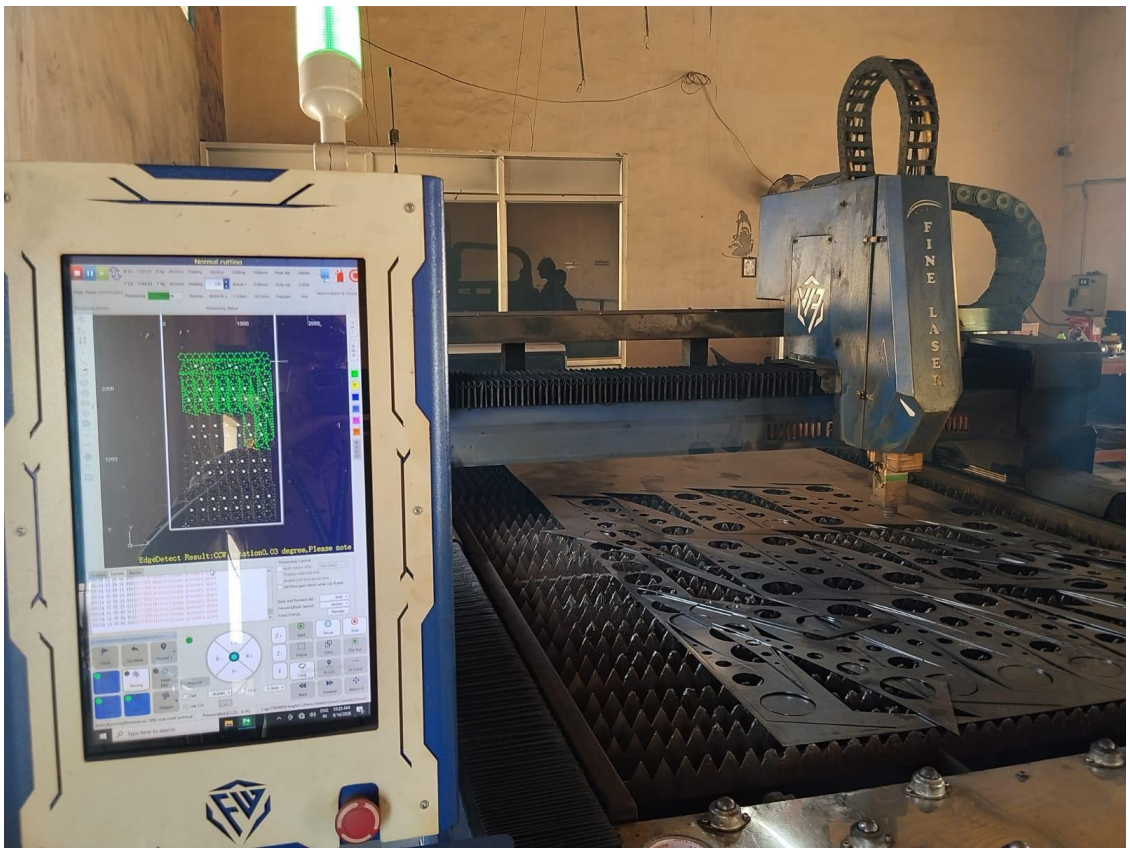


Fig. 1 The laser cutting machine mid-operation, control screen on the left, cutting bed with finished sheet on the right

5. Materials Handled

A few different sheet materials are cut at the unit, depending on what the customer needs:

- **Mild Steel (MS)**, comes as either **Hot Rolled (HR)** or **Cold Rolled (CR)** sheet.
- **Galvanized Iron (GI)** sheet.
- **Stainless Steel (SS)**.
- **Aluminum** / aluminized sheet.

One thing that was explained clearly is the difference between HR and CR mild steel. HR sheet is rolled while the steel is still hot and then shaped, so it comes out a bit rough and is cheaper. CR sheet is rolled cold, without reheating, so the finish is smoother and the dimensions are more accurate, it costs a little more because of that extra precision.

Roughly, the thickness ranges they work with are:

Material	Thickness Range	Remarks
Mild Steel (MS)	0.5 – 20 mm	Cut with Oxygen (O_2)
Stainless Steel (SS)	1 – 10 mm	Cut with Nitrogen (N_2)
GI / Aluminum	1 – 3 mm	Cut with Nitrogen (N_2)

6. Gases Used in Cutting

Something we didn't fully appreciate before the visit is how much the choice of gas matters. Two gases are used, and each one is picked for a reason:

- **Oxygen (O_2)**, used for mild steel. It reacts with the hot metal and actually helps burn through it, which is useful for thicker MS sheets.
- **Nitrogen (N_2)**, used for stainless steel, GI, and aluminum. It doesn't react with the metal, so it keeps the cut edge clean and prevents oxidation/discoloration.

The pressures used were also mentioned to us:

Gas	Pressure	Notes
Nitrogen (N_2)	~14 bar	Stored as liquid nitrogen; pressure builds up in the tank before use
Oxygen (O_2)	~6 bar	Stored as liquid O_2 / in cylinders of about 230 litres

Both gases are kept in liquid form on site, and the pressure is allowed to build up before it's fed to the cutting head, this keeps the gas flow steady during the actual cut. The nitrogen supply lasted approximately 2 days, while the oxygen supply lasted approximately 30 days, with an oxygen capacity of about 230 litres.

Ice could form around a valve because of the low temperature. The gas pressure was allowed to build up before the gas was supplied to the cutting head, and a high-pressure gas flow was needed during cutting.



Fig. 2 Our group with the plant supervisor on the shop floor, laser cutting machine and cut sheet stock visible behind us

7. Software and CNC Programming

The cutting machine runs on software called **Ray2S**, the operator said it's fairly simple to use once you know it. You load in the design, arrange the parts on the sheet so there's as little wastage as possible (called **nesting**), and set your cutting speed, power, and gas pressure from there. Ray2S also allows sheet-metal profiles to be cut according to the programmed design.

Before that stage, the actual design is done in **AutoCAD** and saved out as a DWG or **DXF** file. That DXF file is what gets imported into **Ray2S**, which turns the drawing into **G-code**, the actual set of instructions the machine follows to move and cut. So, the whole path is: draw it in AutoCAD → export as DXF → import into the cutting software → machine cuts it out.



Fig. 3 Control panel of the laser cutting machine, touchscreen, keyboard, and control buttons

8. Power Supply

Since even a tiny power dip mid-cut can ruin a job, the unit runs the machine through an **online UPS** connected via an **MCB panel**. The UPS is backed by a battery, so it keeps supplying steady power (around 13–14 volts at the control side) even if there's a fluctuation from the main line. There's also a **chiller** running alongside the machine to keep the laser source and optics cool, since they heat up during continuous cutting.

In addition to the online UPS and MCB panel described above, the backup could operate for approximately 30 minutes after a power interruption, allowing the machine to be shut down properly. The chiller forms part of the cooling arrangement for the laser cutting machine. A pressure increase due to compression was also existent.

9. MIG Welding

Besides laser cutting, the unit also does **MIG welding** to join parts after they're cut and fabricated. In MIG welding, a **wire electrode** is fed continuously through the welding gun while a **shielding gas** protects the weld from air contamination. It's generally faster and easier to work with than manual arc welding, which is probably why it's the go-to method here for mild steel joints.

10. Quality Check

One small but interesting detail, after the sheet is placed on the bed, the machine runs an **edge-detection** check to see if the sheet is slightly misaligned (they mentioned something like a **0.03-degree** rotation error). It corrects for that automatically before starting the actual cut. It's a good reminder of why **dimensional tolerance** and inspection matter so much in manufacturing, even something this small can throw off a whole batch of parts if it's ignored.

11. What We Learned

Overall, this visit gave us a much clearer picture of:

- How **fiber laser cutting** works and where it's used in industry.
- Why **oxygen** and **nitrogen** are used differently depending on the metal.
- The practical difference between hot-rolled and cold-rolled mild steel.
- Roughly what thickness ranges MS, SS, GI, and **Aluminum** sheets are cut in.
- How a design moves from **AutoCAD** to a finished part, **DXF**, **G-code**, and the machine itself.
- Why a reliable, UPS-backed power supply is a big deal for precision machines.
- The basics of how **MIG welding** is used alongside cutting in fabrication.

12. Conclusion

Overall, the visit to AIRAA Engineering was genuinely useful, a lot more useful than just reading about laser cutting in a textbook. Seeing the machine actually running, hearing why certain gases and materials are chosen, and watching the whole process from a CAD file to a finished cut sheet made a lot of the theory click into place.

It also gave us a better sense of how much precision, the right material choice, and something as basic as a stable power supply all matter in real manufacturing. It was a good, practical addition to what we've been learning in class, and it's given us a better feel for what this side of Mechanical Engineering actually looks like on the ground.

INDUSTRIAL VISIT REPORT 2 -
KIRUTHIKA INDUSTRIES, COIMBATORE

Host Company	Kiruthika Industries, Coimbatore, Tamil Nadu
Date of Visit	14 August 2026 (Friday – AN session)
Department	Mechanical Engineering
Focus Areas	Fiber laser cutting, MIG welding, sheet metal fabrication

1. Introduction

As part of our academic curriculum, we visited **Kiruthika Industries** to gain practical knowledge about modern manufacturing and machining processes. The visit provided us with an opportunity to observe the practical application of concepts studied in Mechanical Engineering, particularly CNC machining, manufacturing processes, metrology, and quality control.

During the visit, we learned about the manufacturing and machining of **motor-pump components**, including components used for connecting pumps and motors. We were introduced to CNC turning, milling, Vertical Machining Centres (VMC), CNC programming, tooling, dimensional tolerances, and inspection methods.

2. Objectives of the Industrial Visit

The main objectives of the visit were:

- To understand practical manufacturing processes used in industries.
- To observe CNC turning and milling operations.
- To understand the working of advanced CNC machines.
- To learn the basics of CNC programming and machine control.
- To understand dimensional tolerance and precision machining.
- To observe measurement and inspection techniques.
- To understand the importance of quality control in manufacturing.

3. Manufacturing and Machining Processes

One of the major areas discussed during the visit was the machining of **motor-pump components**. The pump and motor are connected using suitable components such as adaptors, which require accurate machining for proper alignment and operation.

The manufacturing process involves machining the required component to the specified dimensions followed by inspection. Proper dimensional accuracy is important because errors in the component can affect assembly, alignment, and the performance of rotating machinery.

4. CNC Turning and Milling

We were introduced to **CNC turning**, which is mainly used for manufacturing cylindrical and rotational components. In CNC turning, the workpiece rotates while the cutting tool removes material according to a programmed tool path. Operations such as facing, turning, grooving, boring, and threading can be performed.

We also learned about **CNC milling and Vertical Machining Centres (VMCs)**. Milling uses a rotating cutting tool to remove material from the workpiece. The machines included **3-axis milling and 4/5-axis VMC machining**. Multi-axis machines provide greater flexibility and are useful for machining complex components and surfaces.

The use of advanced CNC machines improves production accuracy and reduces the variation between components.

5. CNC Control and Programming

During the visit, we were introduced to CNC control systems such as **FANUC and Siemens**. The CNC controller interprets the machining program and controls the movement of the machine, spindle, feed, and tools.

A 2-axis CNC turning centre and 3-axis milling were also identified. FANUC was associated with “DX200” and Siemens with “LMW”; these associations have not been independently verified.

We also learned about CNC programming involving the **X and Z axes**, particularly in turning operations. CNC programs contain instructions that control tool movement, spindle operation, feed rate, and machining sequences.

The machine can also accommodate multiple tools through different tool stations. It uses an **8-station tool arrangement**, which allows different machining operations to be performed efficiently without frequent manual tool changes.

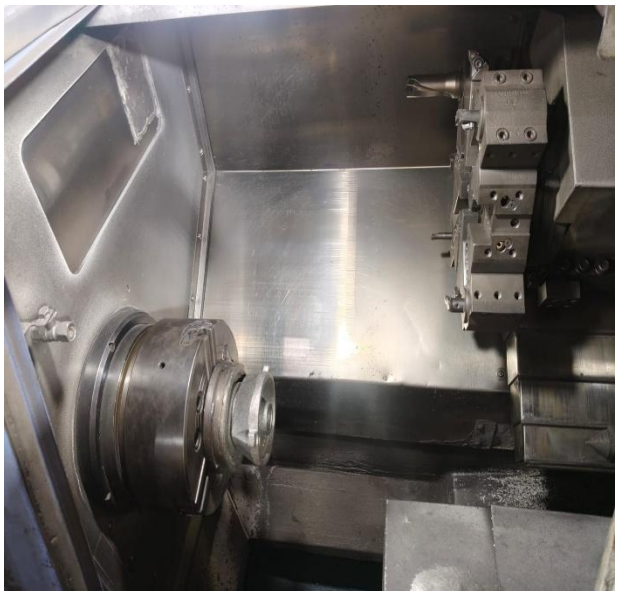


Fig. 1 CNC chuck and tools



Fig. 2 CNC control & program

6. Dimensional Tolerance and Inspection

An important concept observed during the visit was **dimensional tolerance**. In manufacturing, it is practically difficult to produce every component at exactly its nominal dimension. Therefore, a permissible range of variation is provided.

The tolerances observed were approximately **0.02 mm and 0.1 mm**, depending on the component and machining requirement. Such small tolerances demonstrate the importance of precision machining in industrial component manufacturing.

We also learned about the importance of measuring and inspection equipment. Instruments such as **Vernier measuring instruments and dial gauges** can be used to check dimensions, alignment, and small deviations. Precision inspection ensures that the manufactured components satisfy the required specifications before assembly or further processing.

In addition to the tolerances described above, a dimension of 35.00 ± 5 microns was cited, and a bore dial gauge was specifically mentioned for checking tolerance. These details supplement the discussion of dial gauges and dimensional inspection.

7. Materials and Quality Control

The visit also included discussions related to **iron-based/carbon materials** and components associated with industrial applications such as valves and rotating machinery. The selection of material depends on the required strength, durability, wear resistance, and operating conditions.

Casting did not need coolant, which was linked to “loss”; the type of loss was not explained, so this is retained as a brief observation than expanded into a technical claim.

Quality control is an essential part of manufacturing. After machining, components must be inspected to ensure that their dimensions are within the specified tolerance. Proper inspection helps reduce manufacturing defects, assembly problems, material wastage, and machine downtime.

The visit helped us understand the relationship between **machining, dimensional accuracy, measurement, and product quality**.

8. Learning Outcomes

The industrial visit provided us with valuable practical knowledge about modern manufacturing. We gained an understanding of:

- CNC turning and milling operations.
- VMC and multi-axis machining.
- CNC programming and machine control.
- FANUC and Siemens CNC control systems.
- Dimensional tolerance and precision manufacturing.
- Use of measuring and inspection instruments.
- Manufacturing of motor-pump-related components.

The visit also helped us connect classroom concepts from **Manufacturing Processes, Engineering Metrology, CNC Machines, CAD/CAM, and Production Engineering** with their practical applications.

9. Conclusion

The industrial visit to **Kiruthika Industries** was a valuable and informative experience for us as Mechanical Engineering students. It provided practical exposure to CNC machining, motor-pump component manufacturing, turning, milling, VMC operations, CNC programming, tooling, dimensional tolerances, and inspection techniques.

The visit helped us understand the importance of **accuracy, precision, repeatability, quality, and productivity** in modern manufacturing industries. It also gave us a better understanding of how theoretical concepts learned in the classroom are applied in real industrial environments.

Overall, the visit enhanced our technical knowledge and provided useful practical exposure to modern manufacturing technologies, which will be valuable for our future academic studies and careers in Mechanical Engineering.



Fig. 3 Our group outside Kiruthika Industries